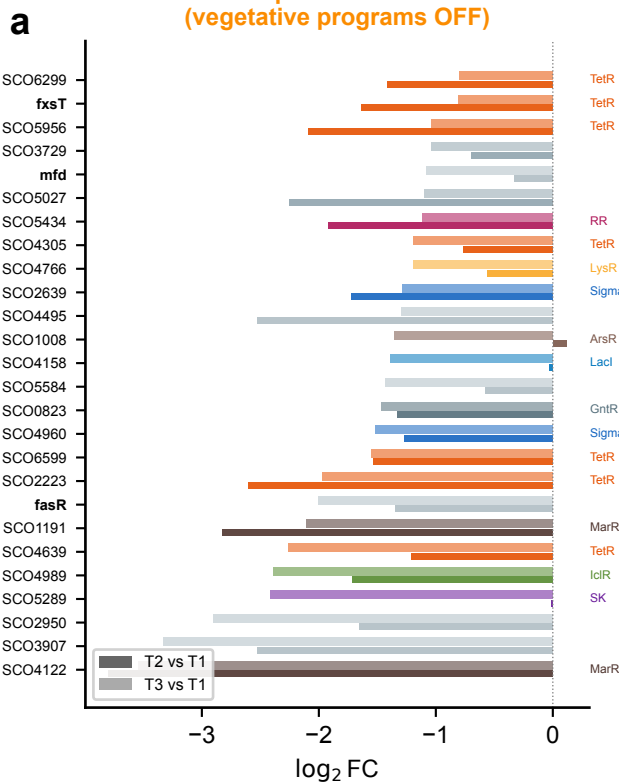
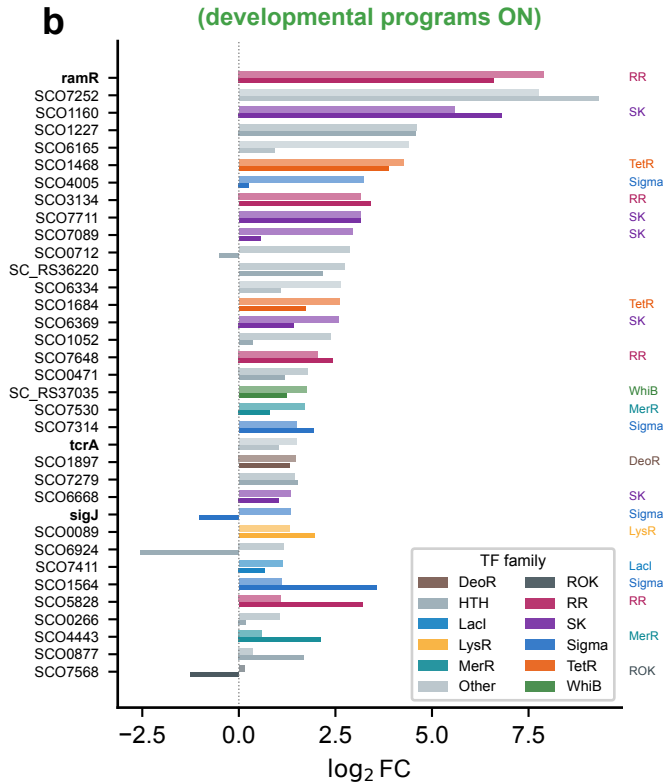


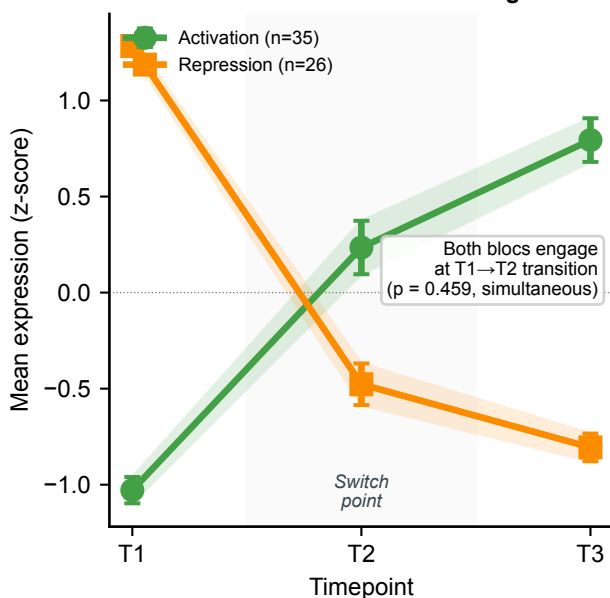
Repression bloc (vegetative programs OFF)



Activation bloc (developmental programs ON)



c Simultaneous switch timing



d Summary of Key Observations

Observation 1

GCCGGC 4mC undergoes geographic redistribution during development
T1: 83% core → T2: 82% arm → T3: residual (n=21)

Observation 2

62 of 1,055 regulatory genes lack methylation protection at TSS
(293 bp threshold, AUC = 0.917; 993 shielded, 62 exposed)

Observation 3

The 62 exposed regulators form two antagonistic blocs
Activation (n=35): TCS, Sigma, WhiB — development ON
Repression (n=26): TetR, metabolic regulators — vegetative OFF
Both blocs switch simultaneously at T1→T2 (MW p = 0.459)

Causation unresolved

Methylation timing is decoupled from expression timing
(p = 0.136, p = 0.30 at T1 → T2; p = 0.012, p = 0.93 at T2 → T3)
This study characterizes methylation distribution and motif patterns;
a causal role for methylation in gene expression regulation remains to be established.